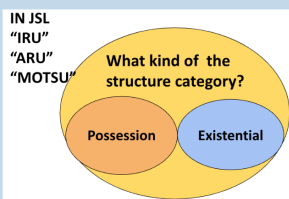


The use of the negative signs DIFFERENT “CHIGAU”, NONE -1 “NAI-1”, and NONE-2 “NAI-2” in Japanese Sign Language

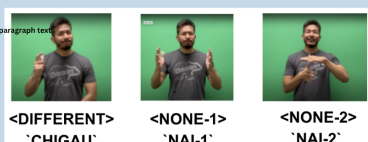
Masashi Tamura Nozomi Tomita Kate Davidson

1. Introduction/Background

- Sign languages have been shown to express existential (There's a cat in the kitchen) and possessive (The kitchen has a cat in it) sentences in varied but related ways that, just as in spoken languages, also connect to expressions of location (in) based on data from American Sign Language (ASL), Austrian Sign Language (ÖGS) and Croatian Sign Language (HZJ) (Chen Pichler et al. 2008).



In JSL, three negative sign can be used with these verbs



The distinct use for IRU and ARU depends on the precedent object (before the Verb), animacy and distance.

EX)

JSL: IX_1 CAT **IRU**

JSL: HOUSE IX **CAT IRU**
CAT ARU X

What are factors that show the distinct use of the negative signs DIFFERENT, NONE, and NONE-2?

Animacy?

The use of Space?

Action vs. State?

Research Question: The aim of this study is to clarify the constraints on the use of specific negation signs in JSL, through interview-based research, to understand how existential and possessive sentences are structured. The study also aims to observe whether spatial expressions, particularly those involving locations (e.g., 'in', 'on', 'at'), make use of space in JSL.

3. Method/

- Qualitative research to gain a detailed understanding of the complex realities surrounding the scope of meaning negative sign and the use of space.
- Participants...4 participants from the JSL Teacher Association (JSLTA)
- Data Collection...Elicit possessive expressions/existential and affirmative/negative sentences.

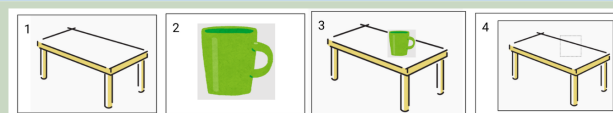
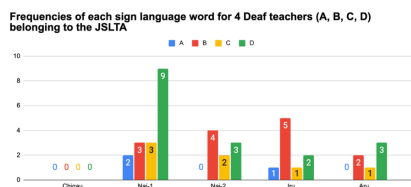
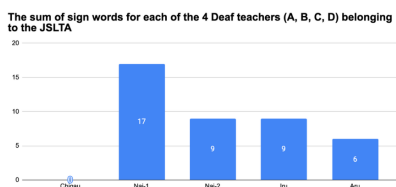


Fig 1

In Zoom...Participants and interviewer shared the google slide (30 pages in total, with 10 questions), Figure 1 represents the flow of signed expressions performed in the order of the slide numbers.

4. Results

Phonological omissions and negative signs in JSL expressions, such as <NONE-1> and <NONE-2>, demonstrate variations in spatial representation depending on conditions of singularity or plurality.



Regarding to the frequency, <DIFFERENT> was absent, while <NONE-1> and <NONE-2> appeared 17 and 9 times, respectively.

	Action or State	Negated existence	Contrastive or conjunctive	Most frequent
DIFFERENT	+	-	-	-
NONE-1	+	+	+	++
NONE-2	-	+	±	+

5. Conclusion

- Proximity was also verified in the papers by De Weedit (2016) and Ichida (2005a).

Due to conditional use...

- Negative signs are observed more frequently than the <IRU> and <ARU> (is/are), likely caused by phonological omission in positive.
- <NONE-1> and <NONE-2> may exhibit different forms, functioning as singular or plural depending on specific contexts.
- <DIFFERENT> was not produced in negative constructions of existential and possessive sentences, contra Ichida (2005).

References:

De Weerd, D. (2016). Existential sentences in Flemish Sign Language and Finnish Sign Language. SKY Journal of Linguistics (Verkkolehti), 29, 7.
Pichler, D. C. Schalber, K., Hochgesang, J., Milkovic, M., Wilbur, R. B., Vulje, M., & Pribanic, L. (2008). Possession and existence in three sign languages. Sign Languages: Spinning and Unraveling the Past, Present and Future. TISLR, 9, 440-458.
市田幸弘. (2005). 「手話の言語学」 (6) 空間の文法—日本手話の文法 (2) 代名詞と動詞の一致. 『月刊言語』 34-6: 90-98. 大修館書店.
Contact
Masashi Tamura: masashi.tamura@gallaudet.edu, Nozomi Tomita: nozomi_tomita@g.harvard.edu, Kate Davidson: kathryndavidson@fas.harvard.edu

Phonological and Spatial [The woman has no paper.] (Fig. 2)

- Spatial Position: The spatial positions of MAN and WOMAN are clearly articulated.
- Phonological Omission: The sign NONE-1 can be performed with either one or two hands.

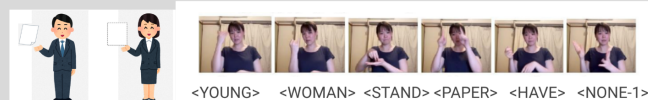
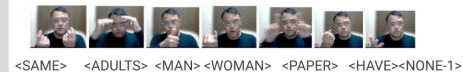


Fig 2



Spatial Constraints [There is a cup but no dish.] (Fig. 3)

- Previous studies (Morgan, 2006) hypothesized that spatial constraints restrict the use of NONE-2. Specifically, in contrastive or concessive sentences, actions or states should avoid using NONE-2.
- However, NONE-2 was observed in use within a contrastive sentence context, contrary to the hypothesis.

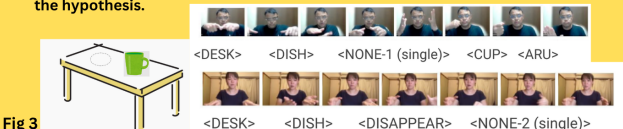
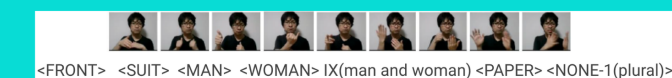


Fig 3

Single and Plural with Spatial expression [A man and woman both do not have paper.] (Fig. 4)

- The expression of negation varies depending on spatial relations and the number of objects being referenced.

Fig 4



Distance in IRU and ARU Sentences[There is a cat in the house.] [There is a cup on the desk.]

- The omission of IRU ("is/are" for nearby objects) and ARU ("is/are" for distant objects) was observed in certain contexts.

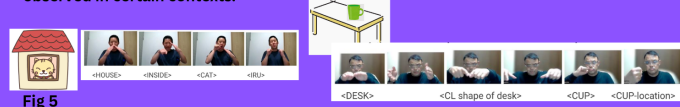


Fig 5